Numerical Methods
LAB 4

1. From today's lecture we have learned about three stopping criteria: Please

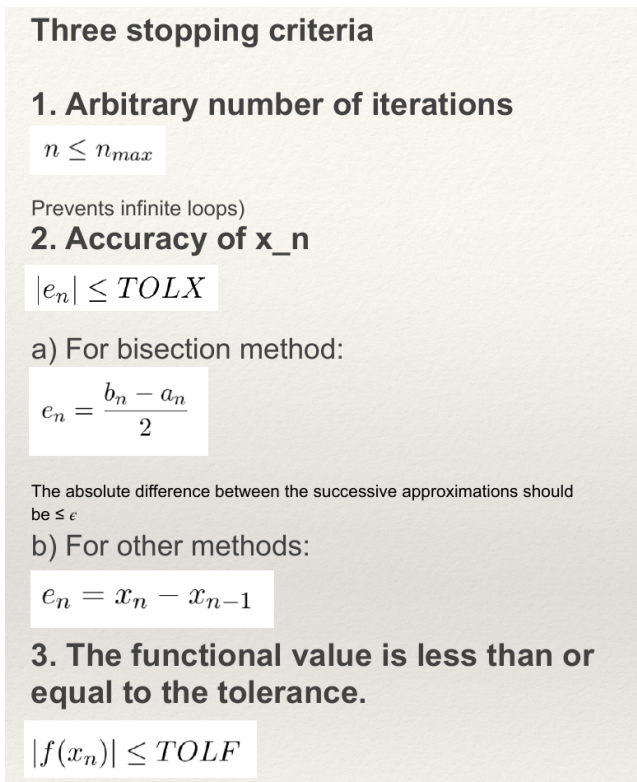

Figure 1: Root finding - stopping criteria

take the python codes of the four root finding methods given on lecture and implement all stopping criteria.

2. We can apply Newton method of root finding to the function $f(x) = x\sin(4x)$ which has many roots.

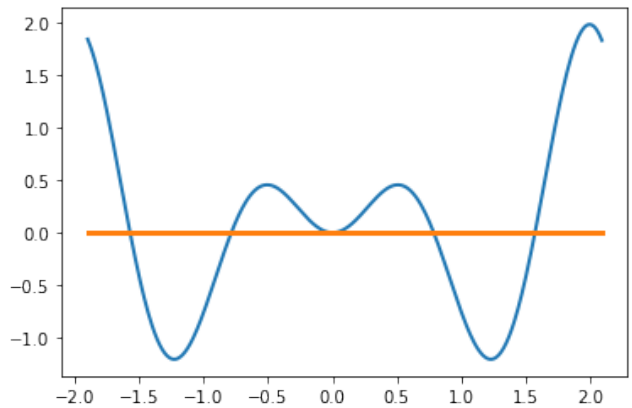

Figure 2: $f(x) = x\sin(4x)$

We can observe, how the root found by the algorithm and the number of iterations depend on the initial guess. The code from from: `https://`

finds a root and plots the function together with the initial guess and the first approximation of a function.

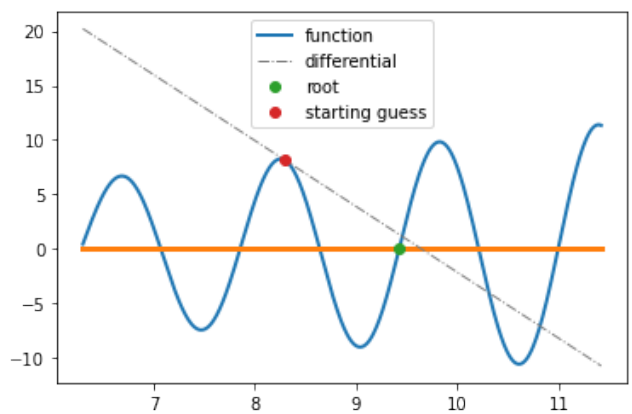

Figure 3: Function $f(x) = x\sin(4x)$ with an initial guess, first approximation of a function and the final root.

Your task is to run this code in a loop, for different initial points and plot, how the root and the number of iterations depends on the initial guess.